

Split Spreadsheet

Split spreadsheet is a Google Sheets spreadsheet that makes splitting the costs between friends simple and quick.

Quick Summary – How to Use the Split Spreadsheet

- 1. Make a Copy**
Use the provided Google Sheets link to create your personal copy.
- 2. Enter Names**
Add names of participants on the People sheet.
- 3. Add Expenses**
Record costs in Expense Sheets. Enter items, amounts, taxes, tips, and discounts as needed.
- 4. Assign Shares**
For each item, specify how much each person owes relative to others in the first sheet.
- 5. Review Totals**
The spreadsheet calculates how much each person should pay or receive. It can also help you minimize the number of transactions.

Following sections will walk you through these steps.

Getting the Spreadsheet File

Make a copy of the spreadsheet file in your personal [Google Drive](#) by clicking on this link:

https://docs.google.com/spreadsheets/d/11TWKigeM_nBcDsNIWsbaQL6YpX9jYOtaxcMP1Uk18Tk/copy



The attached Apps Script file and functionality will also be copied

Google will notify you that the attached Apps Script file and functionality will also be copied. The script is limited to accessing this spreadsheet file and it is safe to copy.

Refer to this document in the future to get the latest update.

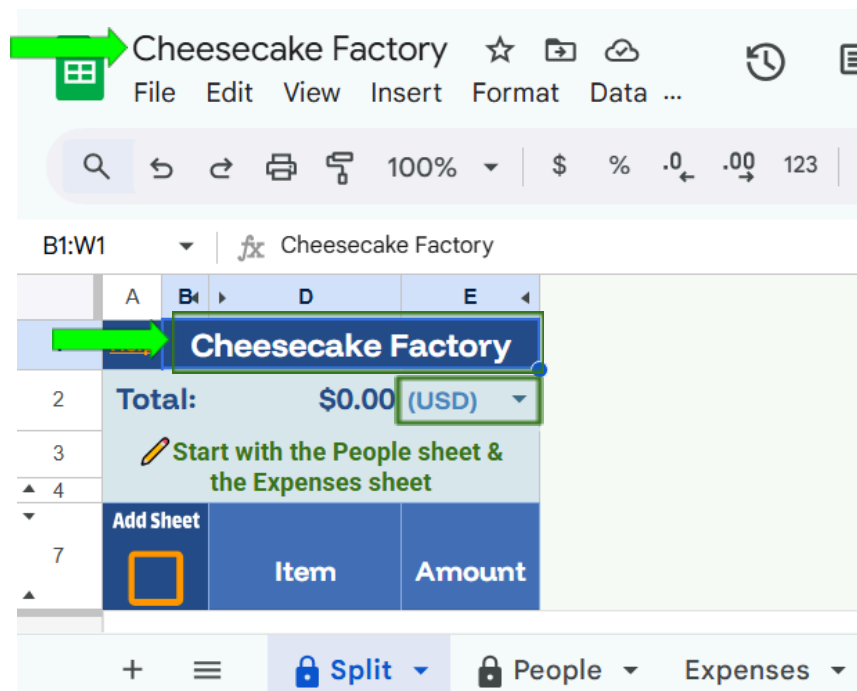
How to Use the Spreadsheet

Start a new split by right clicking on the blank spreadsheet file in your Google Drive and choosing *Make a Copy*. Copying a blank spreadsheet is easier than clearing one you already used.

Let's look at a few examples.

Scenario 1: Split a Restaurant Check

Here, you can modify the title or change the currency. The title and the name of the file are tied programmatically.



Besides the title and currency, you shouldn't touch anything else in the first sheet yet. We will fill out the other sheets and the first sheet will be automatically populated.

 **Not all cells are editable!** In this guide, a green rectangle shows you the editable cells. Do not modify protected cells. Modifying protected cells would result in a warning dialog and you should choose *Cancel*.

Go to the *People* sheet and type in the names of the individuals:

	A
1	Names
2	Jack
3	Chris
4	Rose
5	
6	
7	
8	

+ ≡ **People** ▾

Next, head to the *Expenses* sheet. We usually rename Expense Sheets to the name of the restaurant but since in this example we have only one restaurant check, we're leaving the name.

Go ahead and enter the amounts in the check. Feel free to use formulas if needed:

	A	C
1	Currency:	Default ▾
3	Total:	\$109.23
5	Tax: (6.23%)	\$5.61
6	Tip: (\$13.52)	15%
7	Discount:	\$0.00
8	Item	Amount
10	Beer	\$15.00
11	Pizza	\$19.80
12	Burger	\$17.30
13	Steak	\$32.00
14	Coke	\$6.00
15		
16		

+ ≡ ▾ **Expenses** ▾ < >

If not already done in the check, here we combine the similar items. For example, quantity of beer was 2 (\$7.5 each).

Tax, *Tip* or *Discount* can be entered as amounts or percentage. The title displays the percentage or dollar amount for your information.

We are done here. To start splitting go to the first sheet. It should look like this now:

The image shows two side-by-side screenshots of a mobile application interface for 'Cheesecake Factory'. The left screenshot shows an error state with a red background and the text 'Missing share values'. The right screenshot shows the successful state after splitting, with share ratios assigned to Jack, Chris, and Rose. A green arrow points from the error state to the successful state.

Cheesecake Factory					
Total:		\$109.23 (USD)			
Missing share values					
Add Sheet	Item	Amount	Jack	Chris	Rose
Beer	\$ 15.00				
Pizza	\$ 19.80				
Burger	\$ 17.30				
Steak	\$ 32.00				
Coke	\$ 6.00				

Cheesecake Factory					
Total:		\$109.23 (USD)			
Add Sheet	Item	Amount	Jack	Chris	Rose
Beer	\$ 15.00	1	1		
Pizza	\$ 19.80	1	3	4	
Burger	\$ 17.30	1			
Steak	\$ 32.00		1		
Coke	\$ 6.00		1	1	

The **split section** in this sheet is editable (besides the title and currency mentioned before). The error messages in red show you what's wrong. The issue might be in this sheet, like above, or in another sheet.

Important: Do not delete or reorder any rows/columns here in the Split sheet. Always go back to the Expenses or People sheets. All share assignments will be preserved.

All you can do here is to assign share ratios for each person. Let's see how it's done:

The share ratios are *relative*, meaning that specify the proportion an individual owes toward the item relative to the other individuals in that row. i.e equally, twice, etc...

Let's look at the share amounts entered:

- Jack and Chris each had equal share of the beers. You could put two 2s here and it would mean the same thing. Let's say each had one bottle.
- The pizza had 8 slices. Jack tried 1 slice, Chris had 3 and Rose had 4 slices.
- Jack had the Burger.
- Chris had the Steak.
- Chris and Rose had one coke each (they had equal share of the cokes)
- The tax and tip amounts (shown in small font on the left) are split across everyone based on the share ratios.

We are done! The amount each person's credit card should be charged is over the names.

Scenario 2: Splitting Camping Costs

Splitting the camping costs poses a different challenge since each person paid for different items. After copying the blank file, we rename it to “Camping”, then we fill the People and Expenses sheets:

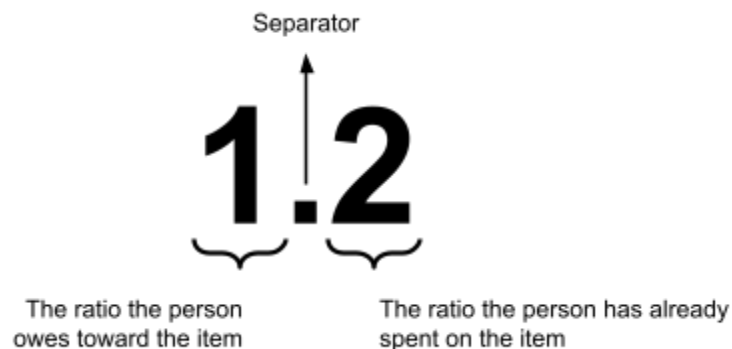
	A
1	Names
2	Jack
3	Rich
4	Emily
5	Jane
6	Bill
7	Chris
8	
9	

+ ≡ **People** ▾

	A	C
1	Currency: Default ▾	
3	Total:	\$272.15
5	Tax:	\$0.00
6	Tip:	\$0.00
7	Discount:	\$0.00
8	Item	Amount
10	Campsite	\$110.00
11	Charcoal	\$9.91
12	Firewood	\$10.00
13	Beef	\$70.00
14	Coke	\$25.20
15	Mosquito Repellent	\$5.04
16	Bread	\$15.00
17	Cheese	\$10.00
18	Eggs	\$10.00
19	Oil	\$7.00

+ ≡ ▾ **Expenses** ▾ < >

We specify the shares now. Specifically, we need to specify two things: who paid for what, and what are the items each person used. This is how we specify two values in one cell:



The splitting looks like this for this camping trip:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	Camping																		
2	Total: \$272.15 (USD)				43.46	(58.95)	12.69	2.48	2.67	(2.35)									
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- Look at the selected cell above. Read it as 1 and 2. This means Emily paid twice as much as Jack for the Mosquito Repellent, and everyone used an equal share. (Emily brought 2 cans, Jack bought another)
- Jack had the beef twice as much as the others. Emily and Chris both bought equal amounts of the beef and brought to the campsite.
- The Cheese and Coke are interesting here since one person bought them for others and didn't use any.

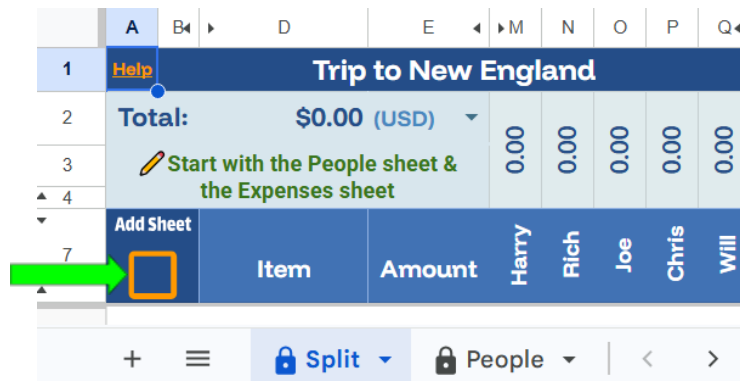
Note: When you specify a payee for an item, you have to go to each row and enter who paid for each item otherwise you will get a warning about missing payee.

We are done! Jack, Emily, Jane and Bill need to pay, and Rich and Chris are owed. Everyone can pay Rich, and he will pay \$2.35 to Chris.

Scenario 3: A Trip with Friends

Let's combine the two scenarios before. This is about a trip that involves multiple checks and types of expenses. Just like the previous scenarios, you will need to enter the names in the People sheet, however, each check and expense type should go into its own Expense Sheet.

To add a new Expense Sheet, click on the *Add Sheet* checkbox:



The screenshot shows a spreadsheet interface. The main table has a header row with columns for 'Item', 'Amount', and five people: 'Harry', 'Rich', 'Joe', 'Chris', and 'Will'. The first row of data shows 'Total: \$0.00 (USD)' with values of 0.00 for each person. A green arrow points to a checkbox labeled 'Add Sheet' in the first column of the table. Below the table, there is a toolbar with a plus sign, a menu icon, a 'Split' button, a 'People' dropdown, and navigation arrows.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Help	Trip to New England															
2	Total:	\$0.00 (USD)															
3		Start with the People sheet & the Expenses sheet															
4																	
7																	

Rename the new Expense Sheet to a meaningful name (e.g., the restaurant's name).

You can delete or reorder the Expense Sheets. When deleting, you get a protected sheet warning. This is the only time you can ignore the warning, and select OK. Don't suppress future warnings.

Note: Avoid reordering or renaming the first two sheets. This is indicated already by  on the sheets.

Let's look at our Expense Sheets after they are created, renamed and filled:

Currency: Default		Currency: Default	
Total: \$1,073.56		Total: \$88.01	
Paid By:		Paid By: Rich	
Tax: \$0.00		Tax: (8.01%) \$5.73	
Tip: \$0.00		Tip: (\$10.73) 15%	
Discount: \$0.00		Discount: \$0.00	
Item	Amount	Item	Amount
Plane Tickets	\$960.56	Burgers	\$24.60
Parking Manchstr	\$32.00	Coke	\$7.00
Icecream	\$30.00	Potato	\$13.95
water	\$11.00	Chickenburger	\$26.00
cash lend	\$40.00		
+ ≡ General McDonalds		+ ≡ Expenses McDonalds	

Currency: Default		Currency: Default	
Total: \$114.06		Total: \$150.24	
Paid By: Will		Paid By: Chris Harry	
Tax: (\$6.06) 6.25%		Tax: (6.25%) \$7.44	
Tip: (16.49%) \$16.00		Tip: (\$23.80) 20%	
Discount: (5.15%) \$5.00		Discount: \$0.00	
Item	Amount	Item	Amount
Empanadas	\$14.00	California Salad	\$15.00
Anticucho	\$13.00	Cacio Pepe Pasta	\$22.00
Ceviche Mixsto	\$19.00	Spicy Rigatoni	\$19.00
Tiradito Exotico	\$17.00	Ahi Tuna	\$26.00
Tiradito Bandera	\$19.00	Peach Lemonade	\$9.00
Sodas	\$15.00	Carne Asada Steak	\$28.00
+ ≡ Peruvian Cheesecake		+ ≡ Cheesecake Factory	

We renamed the Expenses sheet to General, and kept record of all the one-off expenses there.

When additional sheets are added, an extra row labeled **Paid By** is displayed. This lets you specify the people who paid for the whole sheet. If multiple people are selected in Paid By, the cost is split equally between them.

Note: The selected cell shows the discount feature. This will be applied to all the items the Peruvian check proportionally just like tax and tip. In this example, they had a \$5 Groupon.

All the items will be automatically transferred to the first sheet. We have also entered all share values:

- Chris came with his wife. We only added him, which you can see from the two main dishes he ordered in 2 restaurants.
- Row 12: Harry needed some cash to buy something for himself. Joe gave him \$40 cash.
- Different people covered each restaurant's check. This is specified on the Expense Sheets and reflected here by the  background. You could also add .1 on each row, but this way is more convenient.
- Harry and Chris paid equally for the Cheesecake Factory check.

The result is that everyone needs to pay Joe.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Help	Trip to New England															
2	Total:	\$1,425.87 (USD)						132.54	120.37	(783.24)	401.15	129.18					
3																	
4																	
7	Add Sheet		Item	Amount				Harry	Rich	Joe	Chris	Will					
8	General	Plane Tickets	\$ 960.56					1	1	1.1	2	1					
9		Parking Manchstr	\$ 32.00					1.1	1	1	2	1					
10		Icecream	\$ 30.00					1.1	1		1	1					
11		water	\$ 11.00					1	1.1	1	1	1					
12		cash lend	\$ 40.00					1		0.1							
13	McDonalds Tx: 5.73 - Tp: 10.73	Burgers	\$ 24.60								1						
14		Coke	\$ 7.00						1				1				
15		Potato	\$ 13.95					1	1				1				
16		Chickenburger	\$ 26.00					1		1							
17	Peruvian Tx: 6.06 - Tp: 16.00 - Dis: 5.00	Empanadas	\$ 14.00										1				
18		Anticucho	\$ 13.00						1								
19		Ceviche Mixsto	\$ 19.00							1							
20		Tiradito Exotico	\$ 17.00								1						
21		Tiradito Bandera	\$ 19.00								1						
22		Sodas	\$ 15.00										1				
23	Cheesecake Factory Tx: 7.44 - Tp: 23.80	California Salad	\$ 15.00						1								
24		Cacio Pepe Pasta	\$ 22.00								1						
25		Spicy Rigatoni	\$ 19.00										1				
26		Ahi Tuna	\$ 26.00					1									
27		Peach Lemonade	\$ 9.00							1							
28		Carne Asada Steak	\$ 28.00								1						

Additional Features

Multi-Currency Expenses

International trips require entering expenses in different currencies and converting back to a base currency for splitting. Each Expense Sheet can either follow the *default* base currency (as seen in all examples before), or have it's own currency. Here's an example for a road trip that involved 3 currencies. Pay attention to the first two rows of each sheet:

Currency: Default

Total: €152.40

Paid By:

Tax: €0.00

Tip: €0.00

Discount: €0.00

Item	Amount
Gas (OMV Vienna)	€82.00
Parking (City Garage)	€26.00
Highway Vignette	€9.90
Groceries	€18.50
Coffee Stop	€16.00

+ ≡ General-Austria Zum Goldenen Teller

Currency: Hungarian Forint

Exchange Rate: 0.00256 /day

Total: Ft40,200.00

Paid By:

Tax: Ft0.00

Tip: Ft0.00

Discount: Ft0.00

Item	Amount
Gas (MOL Station)	Ft22,000.00
Parking (Budapest)	Ft4,500.00
Highway Toll	Ft5,500.00
Groceries & Water	Ft4,200.00
Snacks	Ft4,000.00

+ ≡ General-Hungary Paprika Bistro

Currency: Serbian Dinar

Exchange Rate: 0.00851

Total: дин.18,200.00

Paid By:

Tax: дин.0.00

Tip: дин.0.00

Discount: дин.0.00

Item	Amount
Gas (NIS Petrol)	дин.10,200.00
Parking (Belgrade)	дин.1,300.00
Toll Road Fees	дин.2,500.00
Groceries & Water	дин.2,400.00
Bakery	дин.1,800.00

+ ≡ Serbia-General Kalemegdan Grill

Currency: Default

Total: €125.48

Paid By: Anna

Tax: €0.00

Tip: (€5.98) 5%

Discount: €0.00

Item	Amount
Wiener Schnitzel	€48.00
Cordon Bleu	€19.50
Tafelspitz	€22.00
Beer	€18.00
Dessert	€12.00

+ ≡ Zum Goldenen Teller

Currency: Hungarian Forint

Exchange Rate: 0.00256 /day

Total: Ft31,400.02

Paid By: Max

Tax: Ft0.00

Tip: (10.19%) Ft3,200.00

Discount: Ft0.00

Item	Amount
Goulash Soup	Ft6,600.00
Paprikash	Ft11,600.00
Grilled Sausage Plate	Ft5,000.00
Beer	Ft5,400.00
Dessert	Ft2,800.00

+ ≡ Paprika Bistro

Currency: Serbian Dinar

Exchange Rate: 0.00851

Total: дин.10,000.08

Paid By: Sophie

Tax: (дин.300.00) 3%

Tip: (8.00%) дин.800.00

Discount: дин.0.00

Item	Amount
Ćevapi	дин.3,400.00
Pljeskavica	дин.1,700.00
Grilled Chicken	дин.1,900.00
Salad & Sides	дин.1,200.00

+ ≡ Kalemegdan Grill

After selecting a non-default currency in an Expense Sheet, a new row will appear with today's exchange rate. If you leave it as is, like the two checks in the middle, the rate updates about once a day, indicated by /day. You can always overwrite it with your preferred value which is also a way to freeze the rate which is indicated by . You can revert to the auto-updating rate by clearing the cell. Exchange rates are obtained from [here](#).

In the Split Sheet, all currencies are converted to and displayed only in the base currency.

Central Europe Trip						
Total:		€701.67 (EUR)	(86.24)	0.12	45.41	40.71
Add Sheet			Lukas	Anna	Max	Sophie
☐	Item	Amount				
General-Austria	Gas (OMV Vienna)	€ 82.00	1.1	1	1	1
	Parking (City Garage)	€ 26.00	1.1	1	1	1
	Highway Vignette	€ 9.90	1	1.1	1	1
	Groceries	€ 18.50	1	1.1	1.1	1
	Coffee Stop	€ 16.00		1		1.1
Zum Goldenen Teller Tip: 5.98	Wiener Schnitzel	€ 48.00	1		1	
	Cordon Bleu	€ 19.50		1		
	Tafelspitz	€ 22.00				1
	Beer	€ 18.00	1	1	1	1
	Dessert	€ 12.00	1			1
General-Hungary	Gas (MOL Station)	€ 56.43	1	1.1	1	1.1
	Parking (Budapest)	€ 11.54	1.1	1	1	1
	Highway Toll	€ 14.11	1	1	1.1	1
	Groceries & Water	€ 10.77	1	1	1.1	1
	Snacks	€ 10.26	2.1		1	
Paprika Bistro Tip: 8.21	Goulash Soup	€ 16.93	1			1
	 Paprikash	€ 29.75		1	1	
	Grilled Sausage Plate	€ 12.82		1		
	Beer	€ 13.85	1	1	1	1
	Dessert	€ 7.18	1	1		
Serbia-General	Gas (NIS Petrol)	€ 86.83	1.1	1	1	1
	Parking (Belgrade)	€ 11.07	1	1.1	1	1
	Toll Road Fees	€ 21.28	1.1	1	1	1
	Groceries & Water	€ 20.43	1.1	1	1	
	Bakery	€ 15.32			1.1	1
Kalemegdan Grill Tip: 2.55 - Tip: 6.81	Ćevapi	€ 28.94	1	1		
	Pljeskavica	€ 14.47			1	
	Grilled Chicken	€ 16.17				1
		€ 15.32		1	1	
	Salad & Sides	€ 10.21	1			1

Debt Settlement Simplification

Most of the time, the simplest way to settle debts is when everyone pays one person who is owed, and that person then pays the others. [If a smaller number of transactions is possible](#), the spreadsheet indicates this by grouping people using different colors. Each group should settle debts within themselves.

In this example, the 7 people are split into two  and  groups to simplify debt settlement.

Debt Simplification									
Total: \$88.55 (USD) ▾			(0.90)	(9.61)	6.52	15.18	(5.67)	(8.61)	3.09
Add Sheet			Henry	Harry	Joe	Dave	Rich	Bill	Jeff
	Item	Amount							
Expenses Tx: 4.96 - Tp: 12.75	BBQ Bacon Burger	\$ 13.95	1.1			1	1		
	Greek Salad	\$ 10.25		1.1			1	1	1
	Mozzarella Sticks	\$ 7.50	1		1.1	1	2		
	Shrimp Tacos	\$ 14.75	1		1		0.1	1	2
	Turkey Club	\$ 12.40	1		2			0.1	
	Margherita Pizza	\$ 11.99				1			1.1

Known Issues

1. Undo functionality doesn't work as expected all the time. You may need to repeat undo many times to undo your changes.
2. The first time you open a new file, it may take up to 10 seconds for the Google Script to initialize. You can begin entering data immediately, but wait 10 seconds for full functionality.
3. Deleting an Expense Sheet will produce an unnecessary warning that can be ignored. Do not check the checkbox in the warning dialog for disabling the warning.
4. Reordering the sheet (by dragging them) requires a temporary edit to any cell for the changes to propagate to the Split Sheet, since Google Sheets does not fire an edit event for this operations.
5. Issues specific to the Google Sheets App on mobile:
 - a. Selecting multiple names for *Paid By* in Expense Sheets is not supported.
 - b. When renaming a file and then opening a new spreadsheet, the title on the first sheet doesn't automatically change until you make a modification.
 - c. In extremely rare cases, the automation in the spreadsheet can stop working. Restart the app completely to resolve the issue.
 - d. The spreadsheet may glitch when used with unreliable internet connection.